

Branch Network Capacity Planning Model v2.1

Retail Banking — Operations Analytics · 2LoD Assessment · May 10, 2026

LOW

EXECUTIVE SUMMARY

The Branch Network Capacity Planning Model v2.1 presents a LOW residual risk profile, characterized by its non-consumer-facing nature, use of interpretable statistical methods (SARIMA), and robust data governance controls. The system operates within established Model Risk Management (MRM) boundaries with a clean validation history and effective human-in-the-loop oversight, though minor gaps in drift monitoring granularity and bias testing documentation require attention.

Escalation Rationale

The model presents a LOW risk profile with no critical control failures. The gaps identified are operational or documentation-related and can be addressed through standard MRM remediation processes without requiring immediate executive escalation.

RISK DIMENSION RATINGS

DATA GOVERNANCE LOW 3 finding(s) · 2 gap(s)	MODEL EXPLAINABILITY LOW 3 finding(s) · 1 gap(s)	BIAS & FAIRNESS LOW 3 finding(s) · 2 gap(s)
HUMAN OVERSIGHT LOW 3 finding(s) · 2 gap(s)	REGULATORY ALIGNMENT LOW 4 finding(s) · 1 gap(s)	INCIDENT & DRIFT MONITORING LOW 4 finding(s) · 2 gap(s)

DIMENSION DETAIL

Data Governance LOW
- Data lineage is fully documented and signed off by the data governance team. - Automated completeness and range checks are in place at ingestion. - Data dictionary is current and missing data imputation is documented. - Documentation of missing data imputation methodology lacks detail on sensitivity analysis for different imputation strategies. - No explicit mention of data retention and archival policies specific to the 2019-2024 training window.

REG: SR 11-7 Section 5 (Model Documentation) · SR 11-7 Section 6 (Data Quality)

Model Explainability LOW
- SARIMA is a fully interpretable statistical model. - Seasonal components, trends, and residuals are documented. - Plain-language forecast narratives with confidence intervals are provided to users. - None identified; the model type inherently supports high explainability.

REG: NIST AI RMF MAP function (Measure and Evaluate)

Bias & Fairness

LOW

- Model operates on aggregate branch-level data with no individual PII.
- No protected class features are used.
- Legal and Compliance confirmed no fair lending or employment law applicability.
- While individual-level bias is not applicable, the assessment does not explicitly address potential geographic or socioeconomic proxy bias in branch-level aggregates, though risk is minimal.
- Documentation of the Legal/Compliance review is referenced but not attached as evidence.

REG: ECOA/Regulation B (Applicability Confirmation)

Human Oversight

LOW

- All recommendations are reviewed and approved by Regional Operations managers.
- Model output is one input among several; managers apply local knowledge.
- Override rate of 12% is within expected range and documented.
- Override reasons are not explicitly stated as being captured in a log for periodic trend analysis.
- No evidence of periodic training or calibration for Regional Managers on interpreting model confidence intervals.

REG: SR 11-7 Section 8 (Model Monitoring)

Regulatory Alignment

LOW

- SR 11-7 applied at proportionate low-risk tier.
- No consumer protection regulatory exposure (ECOA, FCRA, UDAAP).
- NIST AI RMF GOVERN and MAP functions documented.
- Annual independent validation completed with clean opinion.
- None identified; regulatory alignment is appropriate for the risk profile.

REG: SR 11-7 · OCC 2011-12 · FFIEC AI Guidance

Incident & Drift Monitoring

LOW

- Monthly forecast accuracy monitoring with MAPE tracked.
- Drift threshold defined (MAPE degradation >15% triggers review).
- Post-COVID structural break handled via recalibration.
- 24-month clean production history with no incidents.
- Drift monitoring relies solely on MAPE; does not account for distributional shifts in seasonal patterns that may not impact overall MAPE significantly.
- Incident response plan escalation triggers are defined but not tested via tabletop exercise in the last 24 months.

REG: SR 11-7 Section 9 (Model Monitoring) · SR 11-7 Section 10 (Model Validation)

2LOD CHALLENGE QUESTIONS

- Q1** Can you provide the specific sensitivity analysis results for the missing data imputation methodology to demonstrate its robustness under different scenarios?
- Q2** How are override reasons documented and analyzed to ensure that human adjustments are not introducing unmonitored bias or operational inefficiencies?
- Q3** Has the Legal and Compliance review confirming no fair lending applicability been formally signed off and archived for audit purposes?

Q4 What is the rationale for using MAPE as the sole drift metric, and have you evaluated the risk of seasonal pattern shifts that do not significantly impact MAPE?

Q5 When was the last time the incident response plan was tested via a tabletop exercise, and what were the outcomes?

RECOMMENDED ACTIONS

1. Document and archive the formal Legal and Compliance sign-off confirming no fair lending applicability.
2. Implement logging for override reasons to enable periodic trend analysis and ensure human adjustments are monitored.
3. Conduct a tabletop exercise for the incident response plan to validate escalation triggers and response procedures.